

Positional Arguments

When you call a function, Python must match each argument in the function call with a parameter in the function definition. The simplest way to do this is based on the order of the arguments provided. Values matched up this way are called *positional arguments*.

To see how this works, consider a function that displays information about pets. The function tells us what kind of animal each pet is and the pet's name, as shown here:

```
pets.py ❶ def describe_pet(animal_type, pet_name):  
    """Display information about a pet."""  
    print(f"\nI have a {animal_type}.")  
    print(f"My {animal_type}'s name is {pet_name.title()}")  
  
❷ describe_pet('hamster', 'harry')
```

The definition shows that this function needs a type of animal and the animal's name ❶. When we call `describe_pet()`, we need to provide an animal type and a name, in that order. For example, in the function call, the argument 'hamster' is assigned to the parameter `animal_type` and the argument 'harry' is assigned to the parameter `pet_name` ❷. In the function body, these two parameters are used to display information about the pet being described.

The output describes a hamster named Harry:

```
I have a hamster.  
My hamster's name is Harry.
```

Multiple Function Calls

You can call a function as many times as needed. Describing a second, different pet requires just one more call to `describe_pet()`:

```
def describe_pet(animal_type, pet_name):  
    """Display information about a pet."""  
    print(f"\nI have a {animal_type}.")  
    print(f"My {animal_type}'s name is {pet_name.title()}")  
  
describe_pet('hamster', 'harry')  
describe_pet('dog', 'willie')
```

In this second function call, we pass `describe_pet()` the arguments 'dog' and 'willie'. As with the previous set of arguments we used, Python matches 'dog' with the parameter `animal_type` and 'willie' with the parameter `pet_name`. As before, the function does its job, but this time it prints values for a dog named Willie. Now we have a hamster named Harry and a dog named Willie:

```
I have a hamster.  
My hamster's name is Harry.
```
